12 **(a)** Show that the only stationary point on the curve $y = \dfrac{\ln x}{x}$ occurs where $x = \mathrm{e}$, as given in line 45. **[3]**

(b) Show that the stationary point is a maximum. **[3]**

(c) It follows from part **(b)** that, for any positive number a with $a \neq \mathrm{e}$,

$$\frac{\ln \mathrm{e}}{\mathrm{e}} > \frac{\ln a}{a}.$$

Use this fact to show that $\mathrm{e}^a > a^{\mathrm{e}}$. **[2]**